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SRM
INSTITUTE OF SCIENCE & TECHNOLOGY
(Deemed to be University u/s 3 of UGC Act, 1956)

Excel in ROS 2: 3-Day Hands-on Workshop on Autonomous Mobile Robot Navigation

16-18 February, 2026



ROS
Robot Operating System

Organized by

**Department of Electronics and
Instrumentation Engineering**
SRM Institute of Science and Technology
Kattankulathur – 603 203, Tamil Nadu, India

In association with



ABOUT SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

SRM Institute of Science and Technology (formerly known as SRM University) is one of the top ranked Universities in India, offering a wide range of Undergraduate, Postgraduate and Doctoral Programs in Engineering, Medicine & Health Sciences, Management, Science & Humanities, Law, and Agricultural Sciences. SRMIST is accredited by NAAC with the highest Grade of 'A++'. The UGC and MoE, Department of Higher Education, Government of India, granted a graded autonomy status to SRMIST by classifying it as category-I University. SRMIST secured 4th position among private Institutions across our country under ARIIA-2021. QS, the world-renowned international ranking agency, has rated SRMIST as a "4 Star" institute with "5 Star" ratings in Teaching, Employability, Inclusiveness and Online Teaching. SRMIST has been ranked 11th nationally under Universities Category by NIRF (National Institutional Ranking Framework) in 2025.

ABOUT THE DEPARTMENT

The Department of Electronics & Instrumentation Engineering was started in the year 2007-08. The department offers two Undergraduate programs such as B.Tech in Electronics and Instrumentation Engineering, B.Tech in Automation and Robotics and two Postgraduate programs such as M.Tech in Electronics and Control Engineering, M.Tech in Automation and Robotics. The scope of these programs is vast and rapidly expanding to meet the needs of manufacturing and process industries. Our department faculties and students have filed 100+ patents of which 75 have been granted and 96 have been published. The Department has received a total funding of

Rs. 2 crores from agencies like DST, SERB, SPARC, NewGen IEDC etc. The Department is accredited by IET and has been recognized by IASC - Sector Skill Council - Affiliated to Ministry of Skill Development and Entrepreneurship, Govt. of India in 2018 and has established Center of Excellence in Factory automation in collaboration with Mitsubishi Electric India Pvt. Ltd. and Center of Excellence in HMI in collaboration with Schneider Electric India Pvt. Ltd.

ABOUT THE WORKSHOP

This 3-Day Hands-on Workshop on Autonomous Mobile Robot Navigation is designed to provide participants with practical, end-to-end training in ROS 2 (Robot Operating System) and its application to real-world autonomous mobile robots. The workshop covers everything from ROS 2 fundamentals to perception, mapping, and autonomous navigation, ensuring participants gain both conceptual clarity and implementation skills.

Participants will work extensively with ROS 2 workspaces, Python-based node development, robot modeling, simulation, sensor integration, and real robot deployment. The program emphasizes hands-on learning using RViz2, Gazebo, LiDAR, cameras, OpenCV, and embedded platforms such as ESP32, culminating in fully autonomous navigation and testing.

TOPICS TO BE COVERED

- Introduction to ROS2 & Core Concepts
- ROS2 Workspace & Python Programming
- Visualization with RViz2 & TF2
- Robot Modeling (URDF & TF)
- Gazebo and RViz Simulation with ROS2

- Robot Motion Control
- Sensor Integration (LiDAR / Laser Scan)
- Perception using Camera & OpenCV
- System Integration with ROS2, Docker, XLaunch
- Embedded ROS & Real Robot Implementation
- Setup and Environment Preparation For Autonomous Robots
- ESP32 Firmware Upload & Configuration
- Robot Teleoperation & Mapping
- Autonomous Navigation & Testing

RESOURCE PERSON:

Dr. SABIQUE P.V, Founder & CEO, PixelBotix Technologies Pvt. Ltd.,

TARGET PARTICIPANTS

- UG, PG students & Research scholars pursuing research / project work in this field
- Faculty members of Engg. Colleges/ Universities
- Restricted to max. of 40 participants on first-come-first-serve basis

REGISTRATION DETAILS

Registration Fee: Rs. 1200/-

Note: The registration fee includes the workshop kit and refreshments.

LAST DATE FOR REGISTRATION

12th February 2026

REGISTRATION LINK

<https://forms.gle/76SNJeSF6FxF2eMU7>